

FTS Midterm Cheat Sheet

The foundations behind every question, one topic per page



Four topics, built one on top of the other

01	Stationarity	AR(1) · unit roots · cointegration Week 2 · six pages
02	Memory: AR, MA, ARMA	ACF/PACF · Box–Jenkins · forecasts Week 3 · five pages
03	Trends and seasons	ARIMA(p, d, q) · SARIMA Week 4 · two pages
04	Volatility and risk	ARCH · GARCH · EWMA · GJR · VaR Week 5 · six pages

Last two pages: the ten triggers worth memorising

01

Stationarity

Weak stationarity, the AR(1) lens, unit roots and cointegration — Week 2

Stationary means the mean, the variance and the lag links stay fixed

Weak (covariance) stationarity is the working definition from here on

Weak stationarity — three conditions

$$E[X_t] = \mu, \quad \text{Var}(X_t) = \sigma^2, \quad \text{Cov}(X_t, X_{t-k}) = \gamma_k$$

Autocovariance and the ACF

$$\rho(k) = \frac{\gamma_k}{\gamma_0}, \quad \rho(0) = 1, \quad -1 \leq \rho(k) \leq 1$$

Definitions

- Strict: every joint distribution fixed
- Weak: mean, variance, lag-k covariance
- White noise: mean 0, no autocorrelation

Read the picture

- Fast ACF decay → stationary
- Slow, near-one decay → unit root
- Growing swings → not stationary

Stationary means stable statistics, not a flat line

In an AR(1), ϕ decides stationarity, memory and mean reversion

The simplest process that can be stationary or not

AR(1)

$$X_t = c + \phi X_{t-1} + \varepsilon_t, \quad |\phi| < 1$$

Moments when $|\phi| < 1$

$$E[X_t] = \frac{c}{1-\phi}, \quad \text{Var}(X_t) = \frac{\sigma^2}{1-\phi^2}, \quad \rho(k) = \phi^k$$

Three regimes

- $|\phi| < 1$: shocks fade, mean-reverts
- $\phi = 1$: unit root, shocks stay forever
- $|\phi| > 1$: explosive

Memory

- A shock shrinks by ϕ each day
- ϕ near 1: long memory, slow return
- Random walk: variance grows with t

$|\phi| < 1$ fades · $\phi = 1$ never fades · $|\phi| > 1$ explodes

Two kinds of trend need two different cures

Detrend a deterministic trend; difference a unit root — never the reverse

Trend-stationary vs difference-stationary

trend: $X_t = a + bt + u_t$ unit root: $\Delta X_t = \mu + \varepsilon_t$

Over-differencing warning

$$\Delta \varepsilon_t = \varepsilon_t - \varepsilon_{t-1} \Rightarrow \rho(1) = -0.5$$

Order of integration

- $I(0)$: already stationary
- $I(1)$: one difference — most prices
- $I(2)$: two differences — rare

Which cure?

- Shock fades $\rightarrow$ detrend
- Shock stays $\rightarrow$ difference
- Smallest d that makes it stationary

Ask first: does a shock fade or stay?

The ADF test asks whether the lagged level still matters

Unit-root testing turns the picture into a decision

Dickey–Fuller regression (ADF adds the lagged differences)

$$\Delta X_t = \alpha + \gamma X_{t-1} + \sum_{i=1}^p \delta_i \Delta X_{t-i} + \varepsilon_t, \quad H_0 : \gamma = 0$$

Hypotheses

- Null: unit root ($\gamma = 0$)
- Small $p \rightarrow$ reject: stationary
- Large $p \rightarrow$ keep the unit root, difference

Read it carefully

- DF critical values (5% ≈ -2.9), not t
- Low power: near-unit roots slip through
- A level shift can mimic a unit root

Failing to reject is not proof — plot the series and re-test after differencing

Two I(1) series with a stationary spread are cointegrated

Spurious regression, cointegration and error correction

Cointegration

$$X_t, Y_t \sim I(1), \quad Z_t = Y_t - \beta X_t \sim I(0)$$

Error-correction model

$$\Delta Y_t = c + \delta \Delta X_t + \gamma Z_{t-1} + \varepsilon_t, \quad \gamma < 0$$

Spurious vs real

- Two I(1) levels: R-squared and t mislead
- Engle–Granger: regress, ADF the residual
- Differencing is safe but loses the long run

Error correction

- Spread pulls the levels back together
- $\gamma < 0$ sets the speed of correction
- Uses the levels and stays valid

Test the residual, not the R-squared

A pairs trade bets that a stationary spread returns to its mean

Pairs trading — cointegration (Week 2) with volatility-aware bands (Week 5)

Spread and its z-score

$$S_t = Y_t - \hat{\beta}X_t, \quad z_t = \frac{S_t - \bar{S}}{s_S}$$

Recipe

- Find a cointegrated pair (Engle–Granger)
- Enter when $|z| > 2$; exit near 0
- High z : sell rich leg, buy β of cheap

Risks

- A merger or regime change breaks the tie
- Re-test out of sample
- Costs can erase a slow reversion

No stationary spread, no pairs trade — exit when the tie breaks

02

Memory: AR, MA and ARMA

Duality, identification, Box–Jenkins and forecasting — Week 3

AR remembers past values; MA remembers past shocks

Lag operator, AR(p), MA(q) and ARMA(p, q)

ARMA(p, q) in lag-operator form

$$LX_t = X_{t-1}, \quad \phi(L) X_t = c + \theta(L) \varepsilon_t$$

MA(q)

$$X_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

Conditions

- AR stationary: roots outside unit circle
- For AR(1) this is $|\varphi| < 1$
- MA invertible: roots outside too

Memory shapes

- AR: echo shrinks gradually
- MA(q): echo stops after q days
- A finite MA is always stationary

Stationarity and invertibility both mean roots outside the unit circle

A stationary AR(1) is an MA(∞); an invertible MA(1) is an AR(∞)

Unroll one form into the other by substituting again and again

AR(1) $\rightarrow$ MA(∞): today in past shocks

$$X_t = \varepsilon_t + \phi \varepsilon_{t-1} + \phi^2 \varepsilon_{t-2} + \phi^3 \varepsilon_{t-3} + \dots \quad (|\phi| < 1)$$

MA(1) $\rightarrow$ AR(∞): today in past values

$$X_t = \theta X_{t-1} - \theta^2 X_{t-2} + \theta^3 X_{t-3} - \dots + \varepsilon_t \quad (|\theta| < 1)$$

AR(1) $\rightarrow$ MA(∞)

- $\phi = 0.8$: weights 1, 0.8, 0.64, 0.512, ...
- Use: how long a shock lasts
- Needs $|\phi| < 1$: stationary

MA(1) $\rightarrow$ AR(∞)

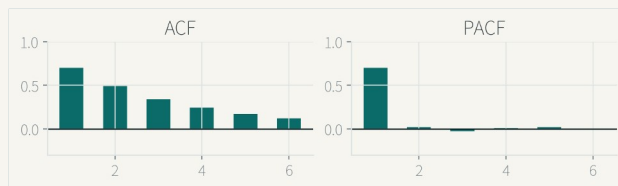
- $\theta = 0.5$: weights 0.5, -0.25, 0.125, ...
- Use: rebuild unseen shocks from data
- Needs $|\theta| < 1$: invertible

One process, two descriptions — choose the one with fewer terms

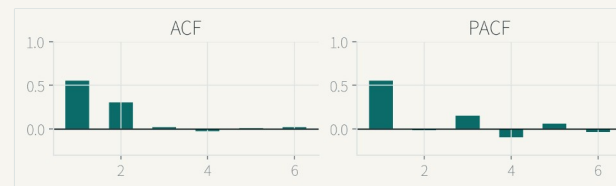
The function that cuts off names the model and its order

ACF = total link (direct + indirect) · PACF = direct link only

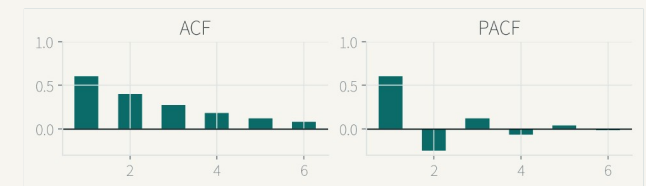
Model	ACF	PACF
AR(p)	tails off gradually	cuts off after lag p
MA(q)	cuts off after lag q	tails off gradually
ARMA(p, q)	tails off	tails off



AR(1): PACF cuts at 1



MA(2): ACF cuts at 2

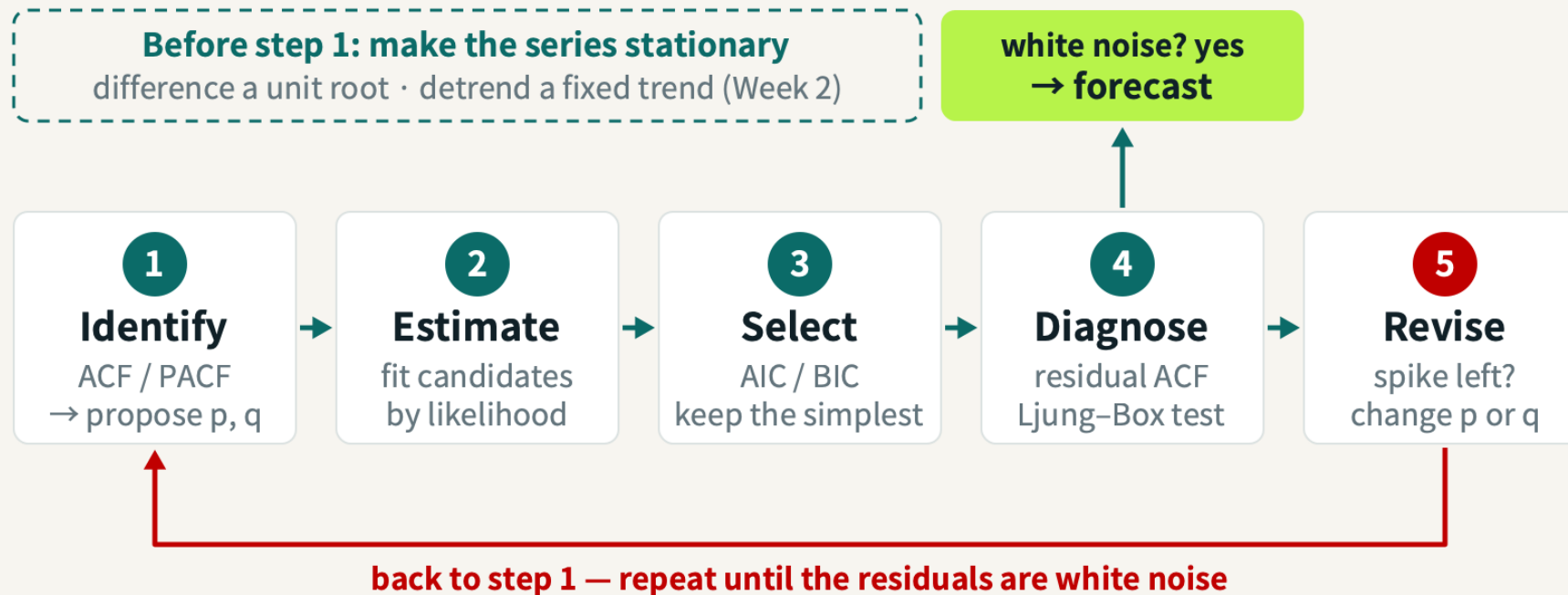


ARMA(1,1): both fade

Both tail off → start with a small ARMA such as (1,1)

Box–Jenkins runs five steps in a loop until the residuals are clean

1 Identify → 2 Estimate → 3 Select → 4 Diagnose → 5 Revise → back to 1



✳ Exit to forecasting only from step 4, when the residuals look like white noise

Steps 3 and 4 use AIC/BIC to select and Ljung–Box to diagnose

Reminder — the two tools inside the Box–Jenkins loop

Step 3 · model selection (lower is better)

$$\text{AIC} = -2\ln L + 2k, \quad \text{BIC} = -2\ln L + k\ln T$$

Step 4 · Ljung–Box test on the residuals

$$Q = T(T+2) \sum_{k=1}^h \frac{\hat{\rho}_k^2}{T-k} \sim \chi_h^2$$

Step 3 · select

- BIC's heavier penalty → simpler model
- When they disagree, lean to BIC
- Parsimony: fewer terms forecast better

Step 4 · diagnose

- Residual ACF should be flat
- Ljung–Box: a large p is good news
- Spike left → step 5, revise p or q

Only white-noise residuals let you leave the loop and forecast

03

Trends and Seasons

ARIMA(p, d, q) and SARIMA — Week 4

ARIMA differences d times, fits an ARMA, then integrates back

The I is just the number of differences, d

ARIMA(p, d, q)

$$\phi(L) (1 - L)^d X_t = c + \theta(L) \varepsilon_t$$

Unit-root forecasts carry on

RW with drift : $\hat{X}_{t+h} = X_t + h\mu$, $\text{Var}(e_h) = h\sigma^2$

Reading (p, d, q)

- p AR terms · d differences · q MA terms
- ARIMA(0,1,0): the random walk
- ARIMA(p,0,q): a stationary ARMA

Choosing d

- ADF on the level, difference, re-test
- Stop at the smallest d that works
- Too many: -0.5 at lag 1, bigger variance

ARMA forecasts revert; ARIMA forecasts continue the trend

Seasonality is memory at lag s, and SARIMA applies ARIMA twice

Seasonal differencing and SARIMA(p, d, q)(P, D, Q) with period s

Seasonal difference (monthly data)

$$(1 - L^{12}) X_t = X_t - X_{t-12}$$

SARIMA

$$\Phi(L^s) \phi(L) (1 - L)^d (1 - L^s)^D X_t = \Theta(L^s) \theta(L) \varepsilon_t$$

Spot it

- ACF spikes at s, 2s, 3s
- s = 12 monthly, 4 quarterly, 7 daily
- Swings grow with level → take logs

Model it

- d for the trend, D for the season
- Airline: (0,1,1)(0,1,1), s = 12
- No seasonal spikes left in residuals

A regular difference removes a trend; a seasonal difference removes a season

04

Volatility and Risk

ARCH, GARCH, extensions and Value-at-Risk — Week 5

Returns are unpredictable in direction but predictable in size

Conditional variance and the ARCH model

Size times direction

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \sim (0, 1)$$

ARCH(1) and its long-run variance

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2, \quad \bar{\sigma}^2 = \frac{\omega}{1 - \alpha}$$

Conditional vs unconditional

- Unconditional: one long-run number
- Conditional: today's, given recent days
- Up after big shocks, down in calm

ARCH facts

- Shock is squared: sign drops out
- Clustering and fat tails for free
- Test: squared-return ACF, ARCH-LM

Uncorrelated returns with correlated squared returns = ARCH effects

In GARCH(1,1), α reacts to news and β carries the memory

Persistence $\alpha + \beta$ sets how long a volatility shock lasts

GARCH(1,1) and its long-run variance

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta}$$

Half-life of a volatility shock

$$h_{1/2} = \frac{\ln 0.5}{\ln(\alpha + \beta)} : \quad 0.95 \Rightarrow 13.5, \quad 0.99 \Rightarrow 69 \text{ days}$$

Persistence $\alpha + \beta$

- 0.95: about 2–3 weeks to half
- 0.99: over two months to half
- Must be below 1 (= 1 is IGARCH)

Extensions

- Student-t: fatter tails (lower ν)
- Asymmetry: GJR-GARCH (two pages on)
- One β replaces many ARCH lags

The closer $\alpha + \beta$ is to 1, the longer a storm of risk lingers

EWMA is a GARCH(1,1) with no constant and no mean reversion

RiskMetrics volatility — one parameter λ , nothing to estimate

EWMA variance (RiskMetrics)

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2, \quad \lambda = 0.94 \text{ (daily)}$$

The same as GARCH(1,1) with

$$\omega = 0, \quad \alpha = 1 - \lambda, \quad \beta = \lambda \Rightarrow \alpha + \beta = 1$$

Like GARCH

- λ plays β (memory), $1 - \lambda$ plays α (news)
- Recent squared returns weigh most
- $\lambda = 0.94$: weight half-life ≈ 11 days

In practice

- RiskMetrics default for daily VaR
- Fast, robust, no estimation step
- No long-run level ($\alpha + \beta = 1$)

Use EWMA for quick daily risk; use GARCH when a long-run level matters

GJR-GARCH lets a fall raise volatility more than an equal rise

Volatility asymmetry, also called the leverage effect

GJR-GARCH(1,1): $I = 1$ after a fall, 0 after a rise

$$\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1}) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

Effect of yesterday's shock on today's variance

$$\text{fall: } (\alpha + \gamma) \varepsilon_{t-1}^2 \quad \text{rise: } \alpha \varepsilon_{t-1}^2$$

Why asymmetry

- Equities: bad news lifts vol more
- Leverage: a fall raises debt/equity
- Plain GARCH squares the shock: blind to sign

Reading GJR

- $\gamma > 0$ measures the leverage effect
- News impact curve: steeper on the left
- EGARCH: same idea in log variance

In ARCH and GARCH only the size matters; in GJR the sign matters too

GARCH bands let pairs-trading thresholds breathe with the spread

Pairs trading, Stage I — fit a GARCH to the spread's changes and scale the bands

GARCH on the spread's changes

$$\sigma_{S,t}^2 = \omega + \alpha (\Delta S_{t-1})^2 + \beta \sigma_{S,t-1}^2$$

Entry and exit bands

$$\text{bands}_t = \bar{S} \pm k \sigma_{S,t} \quad (k \approx 2)$$

Fixed bands

- Too tight in storms → false entries
- Too wide in calm → missed reversions
- Assume the spread's risk is constant

GARCH bands

- Widen in volatile regimes
- Tighten in calm regimes
- Fire only when a move is unusual today

Same z-score idea, measured against today's spread volatility

A 99% VaR is a threshold crossed on about 1% of days

Value-at-Risk, Expected Shortfall and back-testing

VaR from a GARCH volatility (normal quantile)

$$\text{VaR}_{99,t} = -\mu + 2.33 \sigma_t \quad (95\% : 1.645)$$

Expected Shortfall and expected breaks

$$\text{ES} = E[\text{loss} \mid \text{loss} > \text{VaR}], \quad \text{breaks} \approx (1 - c) N$$

Read it

- A threshold, not the worst case
- 250 days → about 2–3 breaks
- ES looks into the tail (Basel)

Back-test it

- Count breaks: Kupiec test
- Check that breaks do not cluster
- Fixed VaR breaks bunch in storms

A good VaR: the right number of breaks, scattered in time

Ten triggers worth memorising (1/2)

If the question shows this, think that

If you see...

Level drifts · ACF decays slowly near 1

Big negative lag-1 correlation after a difference

Two wandering series, gap keeps returning

ACF cuts off after lag q

PACF cuts off after lag p

Think...

Unit root → difference

Over-differenced → one fewer d

Cointegration → spread, ECM, pairs trade

MA(q)

AR(p)

Check stationarity before anything else

Ten triggers worth memorising (2/2)

If the question shows this, think that

If you see...

ACF and PACF both tail off

ACF spikes at 12, 24, 36

Returns uncorrelated, squared returns correlated

$\alpha + \beta$ close to 1

VaR breaks bunch in one crisis

Think...

ARMA — start with (1,1)

Season $s = 12 \rightarrow$ seasonal difference

Volatility clustering $\rightarrow$ ARCH/GARCH

Long half-life, slow mean reversion

Fixed VaR — use a GARCH VaR

Direction has no memory; size does

Stationary first

Then read the memory

Then model the risk

**Know the rule behind
every formula on these pages**

FTS midterm cheat sheet · built from the Week 2–5 lecture decks (Stationarity · Unit Roots · Cointegration · AR/MA · ARMA Identification · Diagnostics & Forecasting · ARIMA · SARIMA · ARCH · GARCH · VaR)